Towards Secure Container Infrastructure on RISC-V: The Development from RustVMM to Kata-Containers

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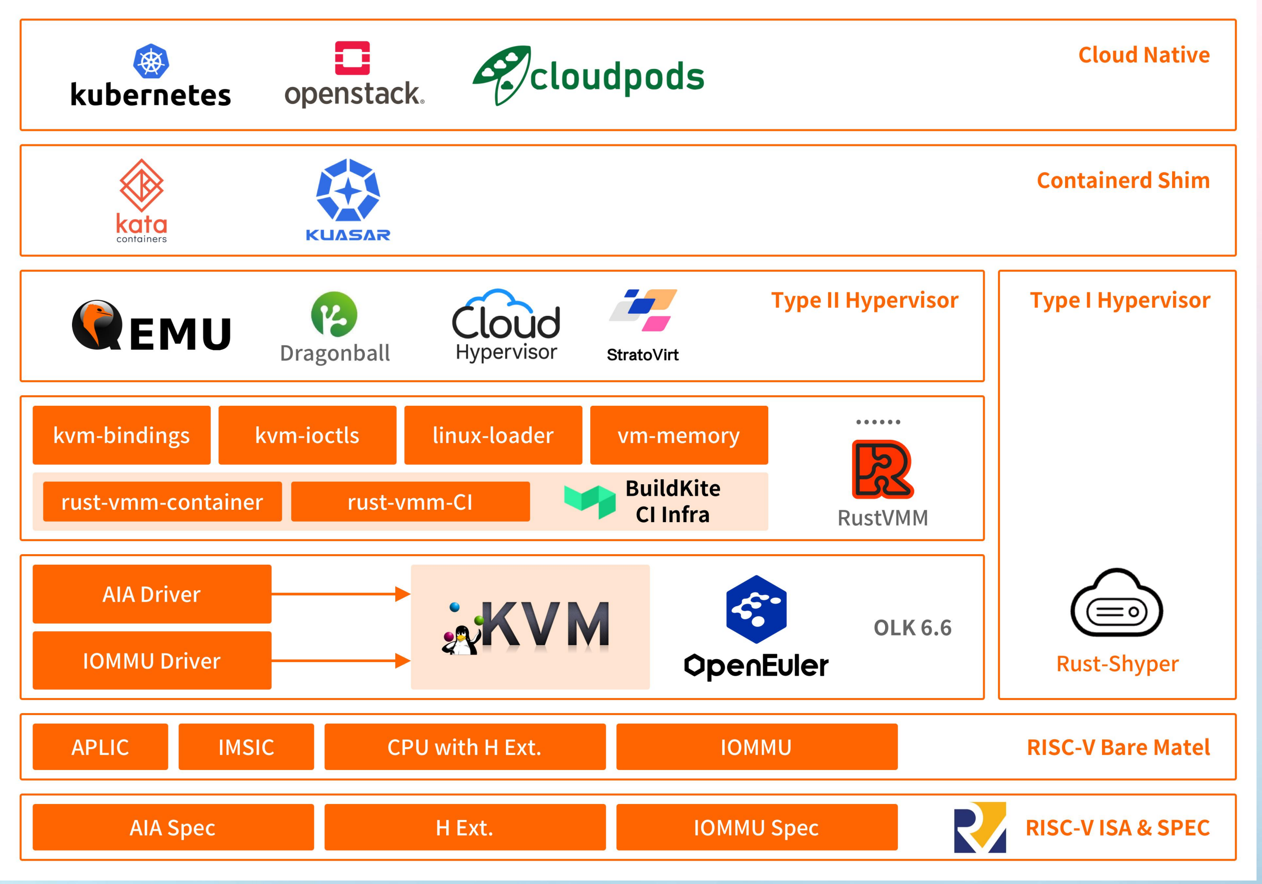
Roadmap — Stage One

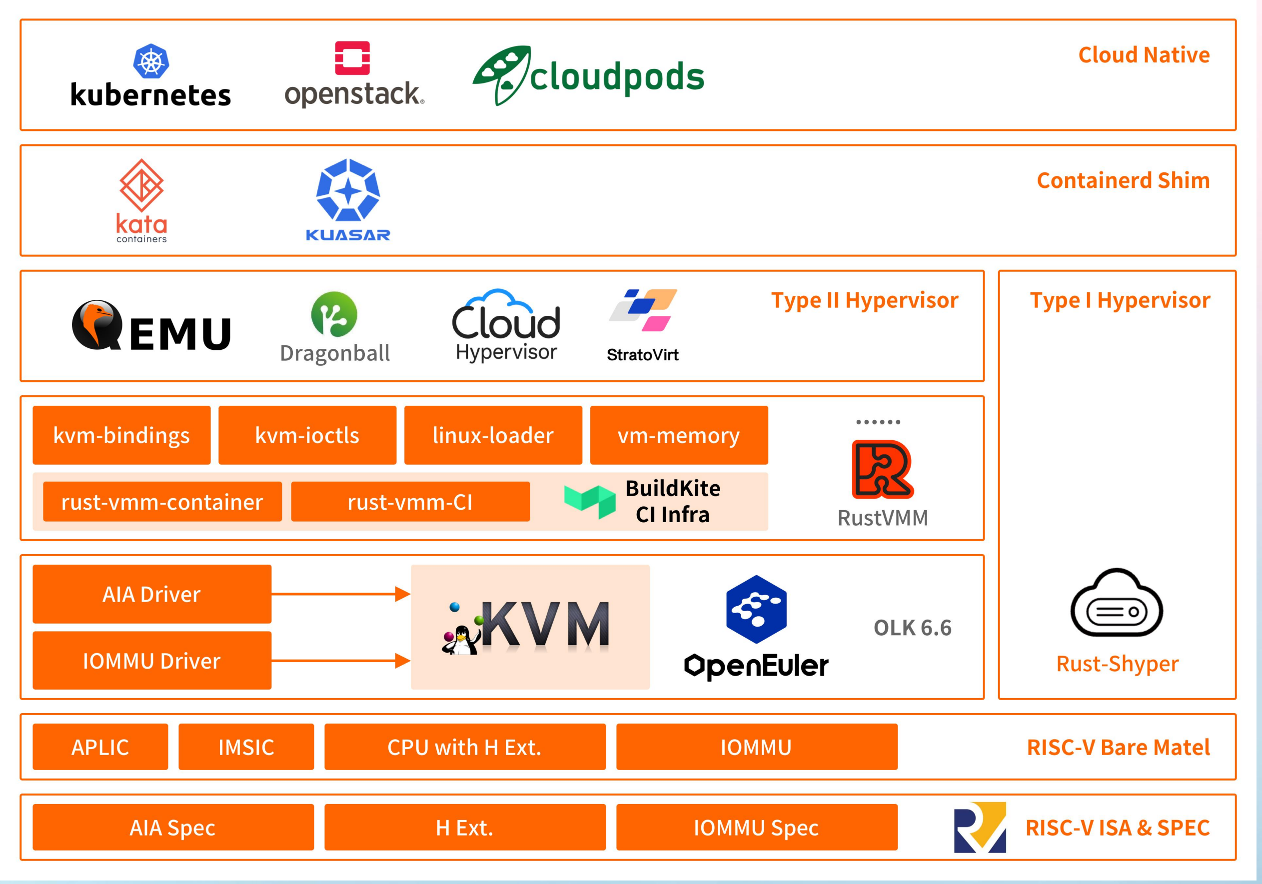
A bottom up, usable, maintainable and dependable software stack on RISC-V platforms.

- Orchestrated by Kubernetes
- Secured by Kata-Containers
- Speedup by Cloud-Hypervisor
- Powered by RustVMM
- On openEuler RISC-V



Roadmap — Stage Two

A fully supported, reliable and prosperous virtualization software ecosystem of RISC-V.
Enabling RISC-V to compete with x86 and ARM in both Virtualization & Cloud Native software ecosystem.



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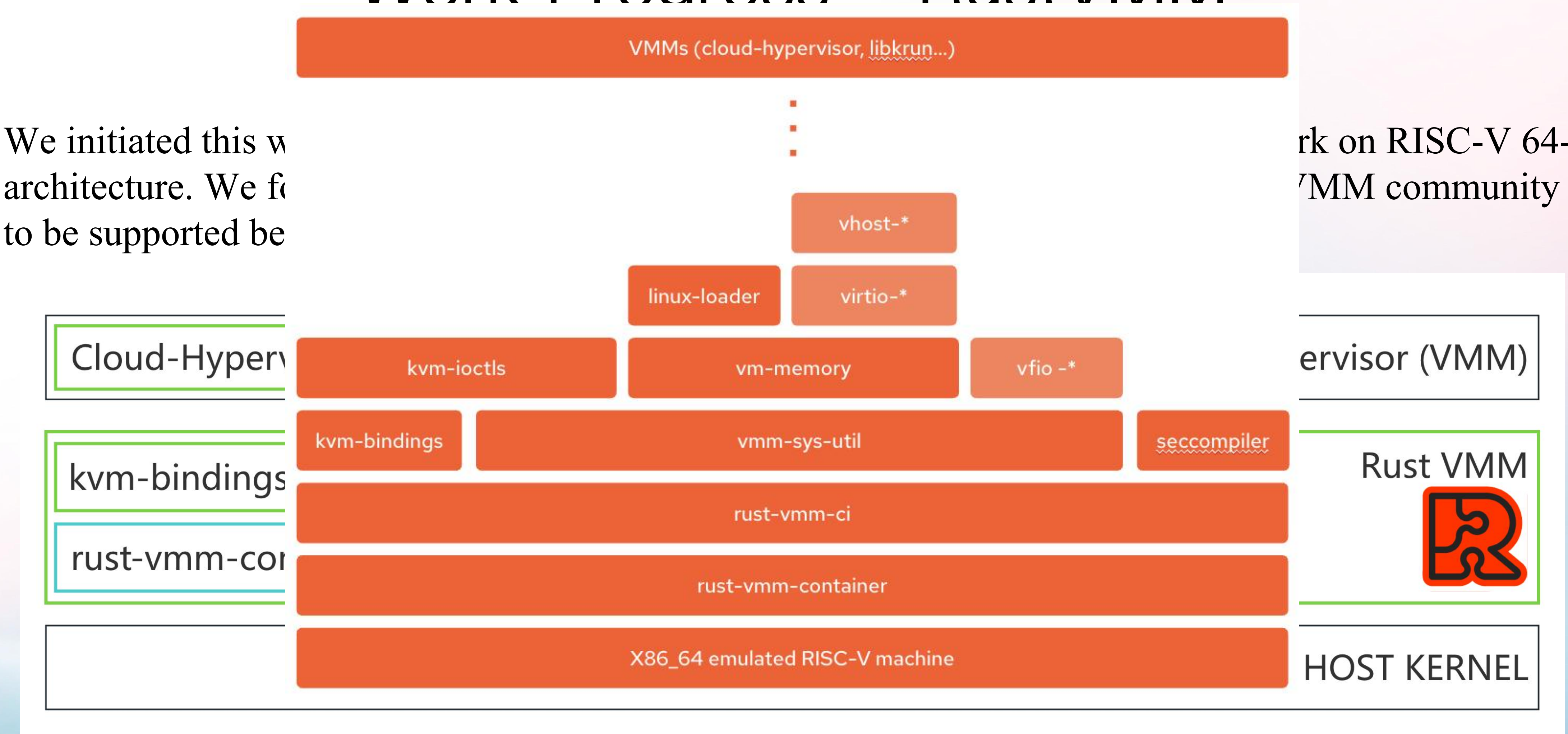
02 Work Progress

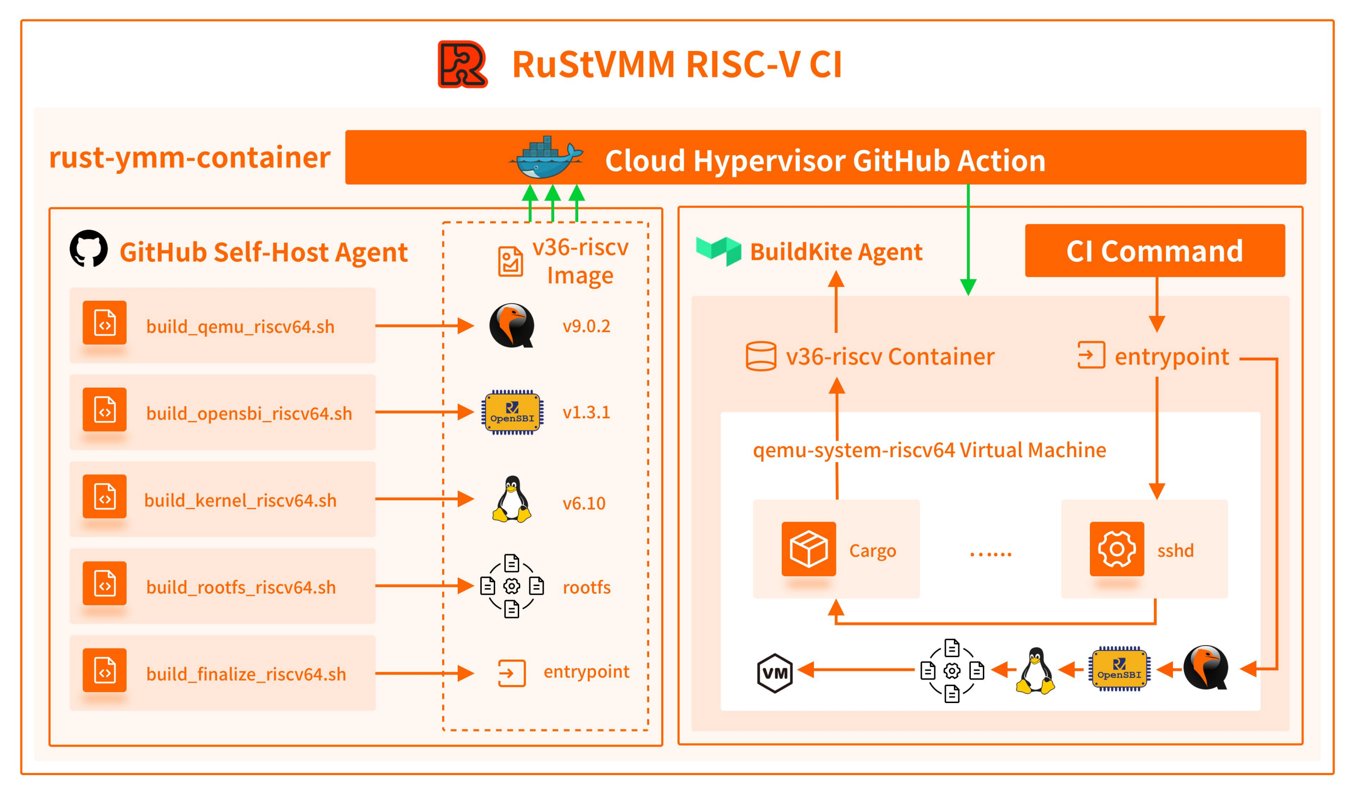
Work Progress – RustVMM

We initiated this w rk on RISC-V 64-bit
architecture. We f MM community need
to be supported be



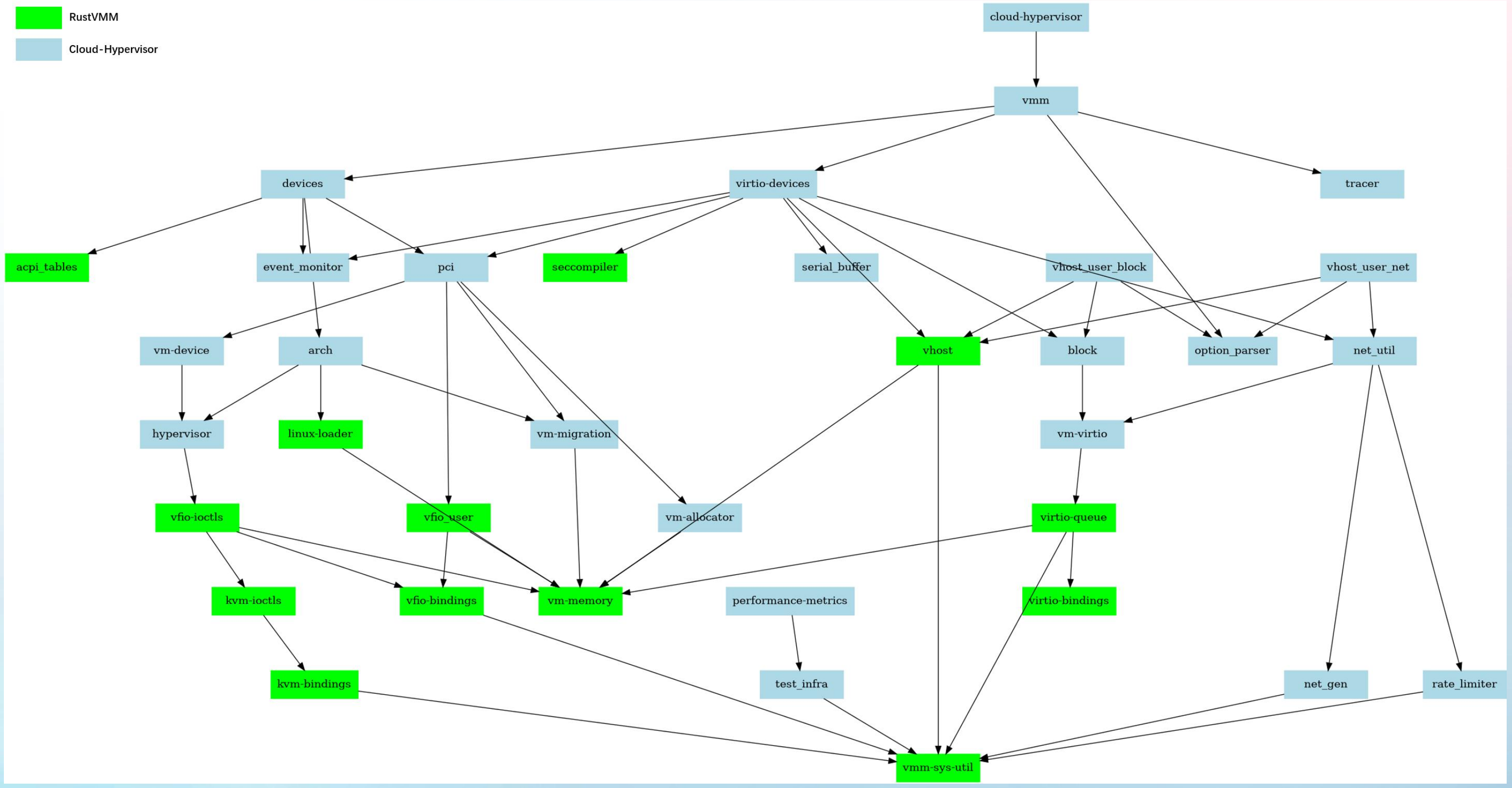
Work Progress – RustVMM

As aforementioned that the actual hardware is not yet made available for us, how the CI should be designed to look after RISC-V code becomes a problem.

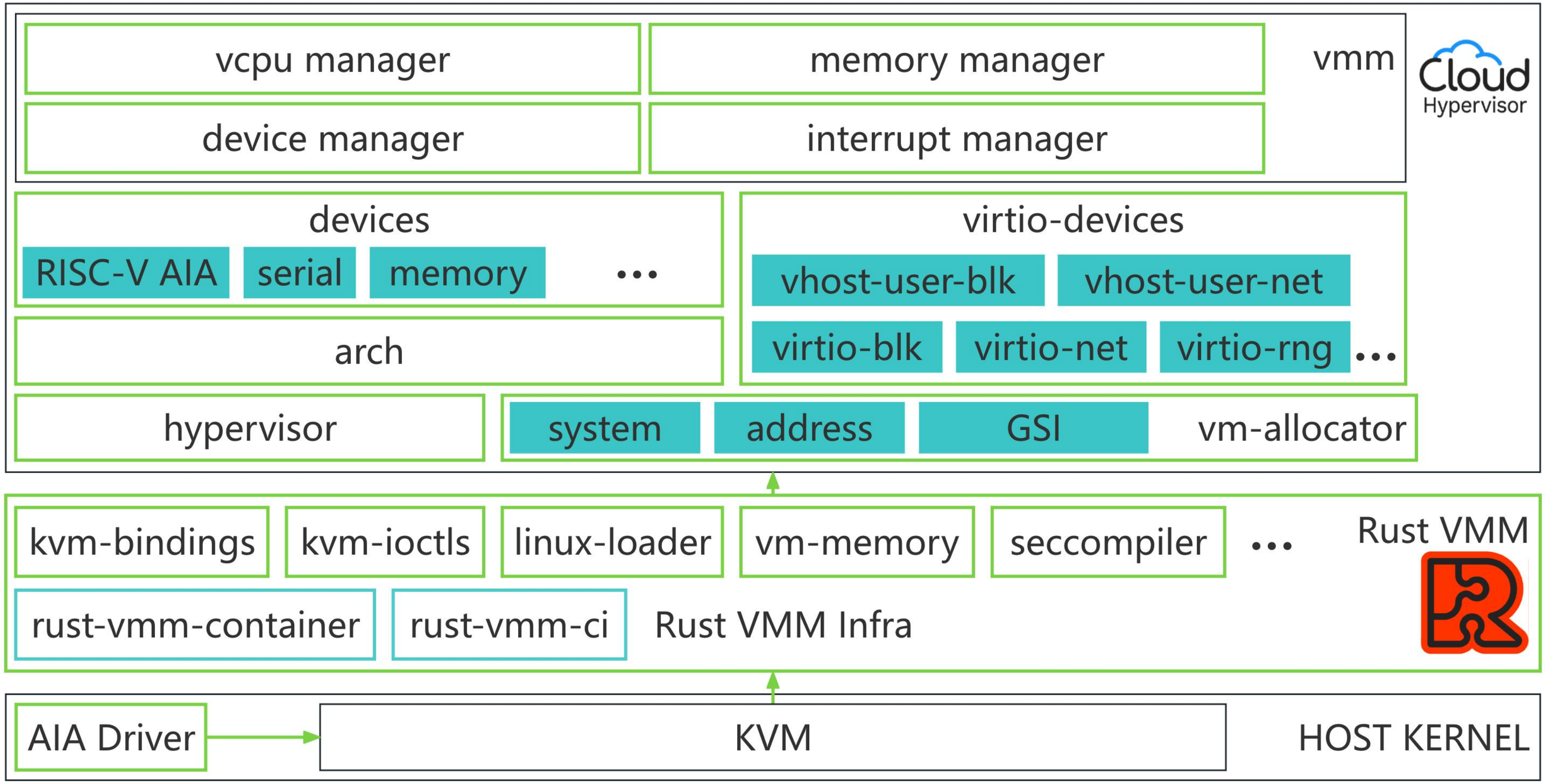


Work Progress – Cloud-Hypervisor

This is a dependency graph of Cloud-Hypervisor, which shows the inter-connection of crates of Cloud-Hypervisor and RustVMM.

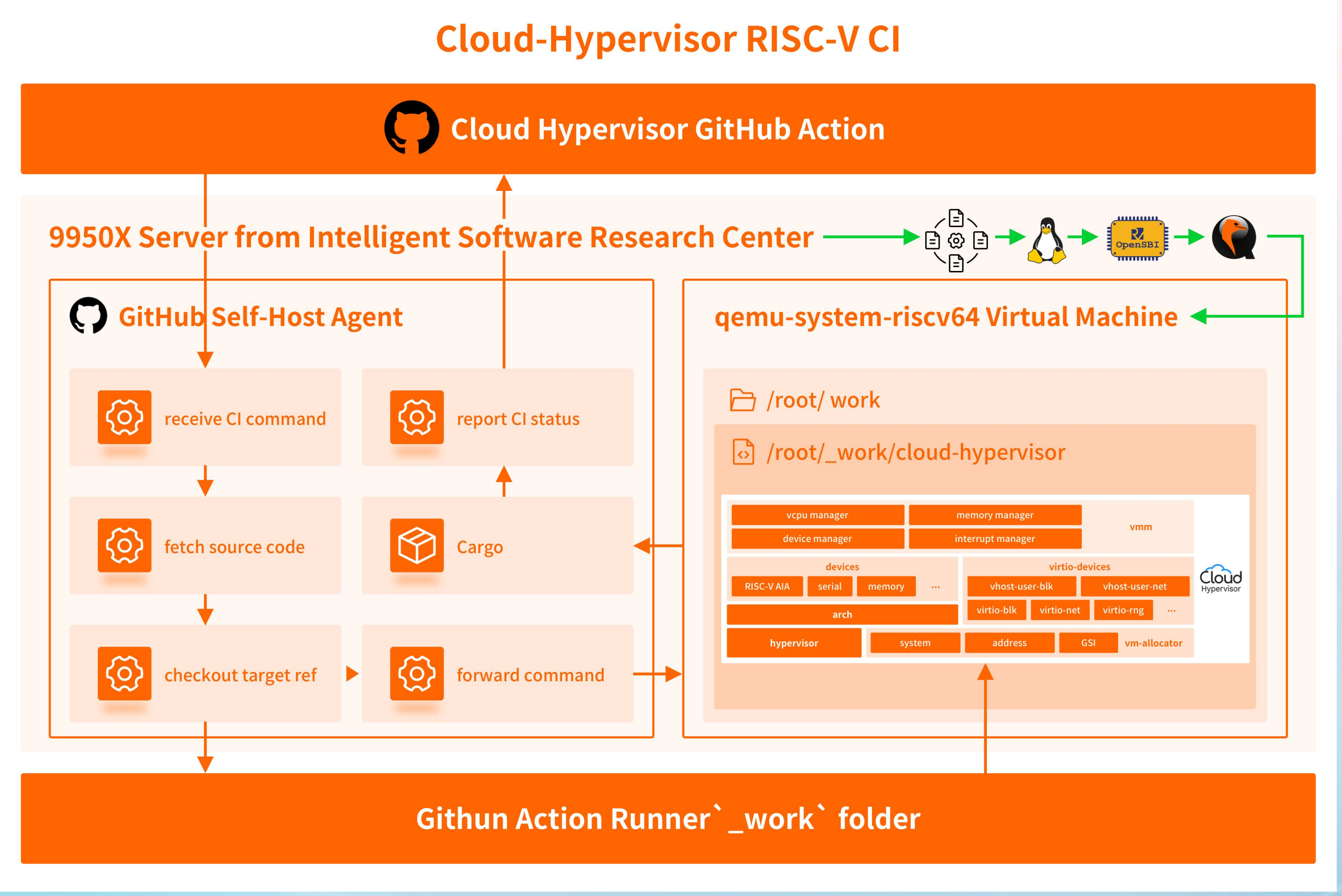


Stage 1 RISC-V Support — Cloud-Hypervisor



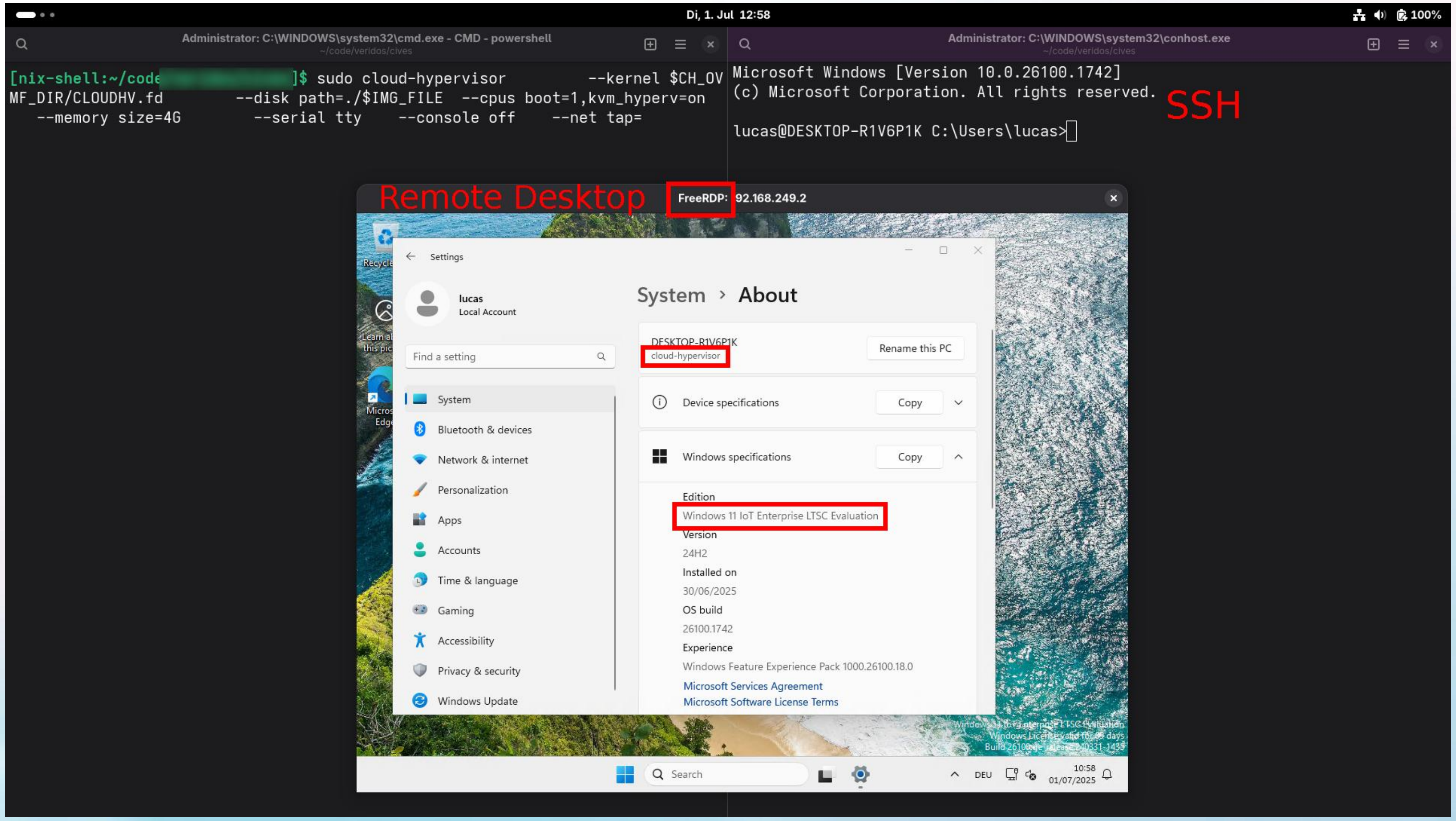
CI Status — Cloud-Hypervisor

Here again, CI for H extension based software like Cloud-Hypervisor need special design and handling. We adopted similar approach used in RustVMM community.



Stage 2 RISC-V Support – Cloud-Hypervisor

Cloud-Hypervisor v45 and onwards now supports direct boot of RISC-V Linux, but there are still many feature gaps.



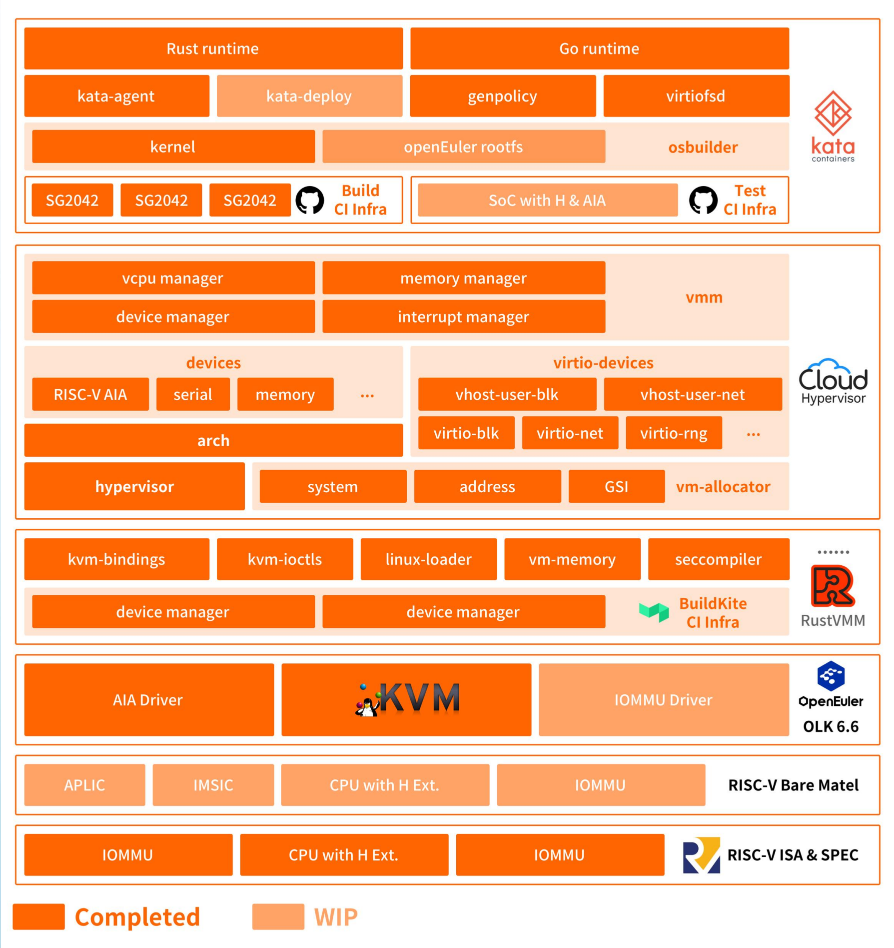
Work Progress — Kata-Containers

We have now worked out a full Rust-based secure container software stack, RISC-V support for:

- Go runtime
- Rust runtime
- Agent

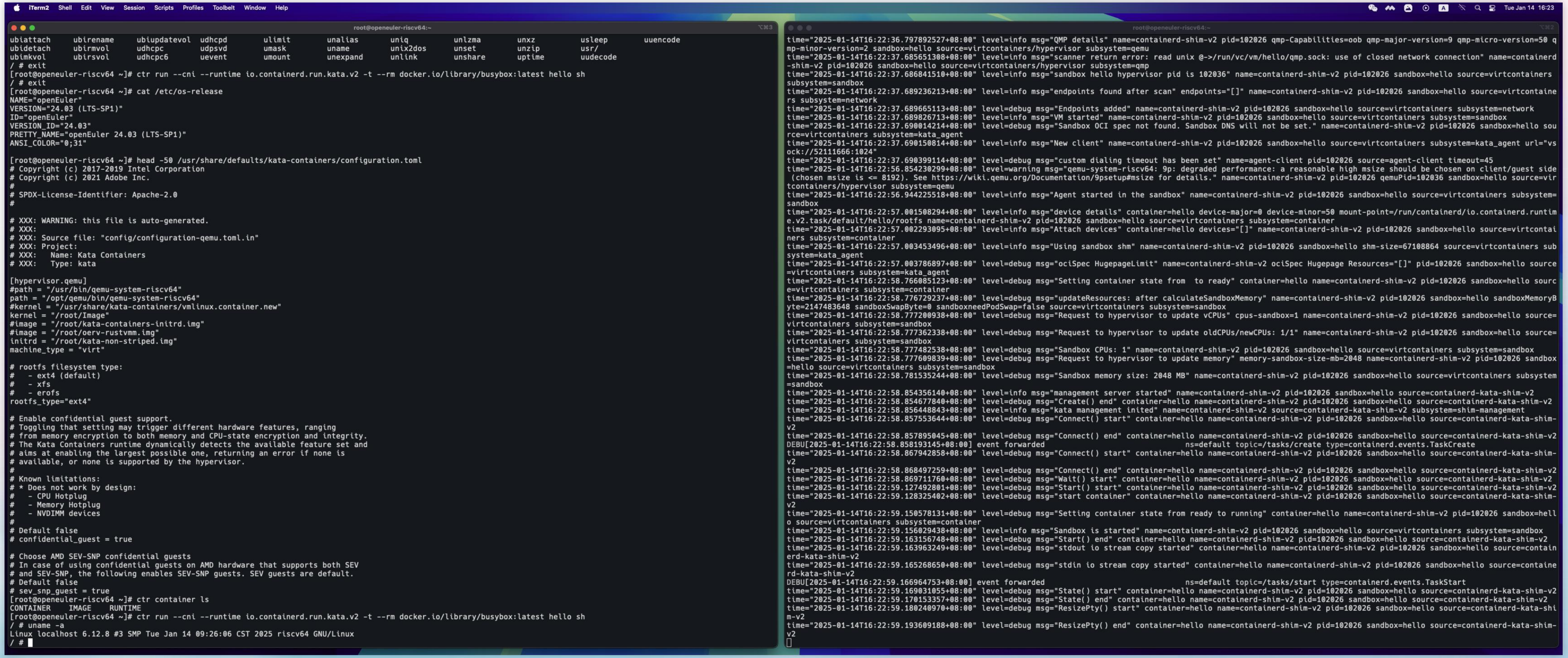
are completed, released in 3.17.0.

Based on RISC-V CI Infra for Kata, with real RISC-V hardware (SG2042) for building.



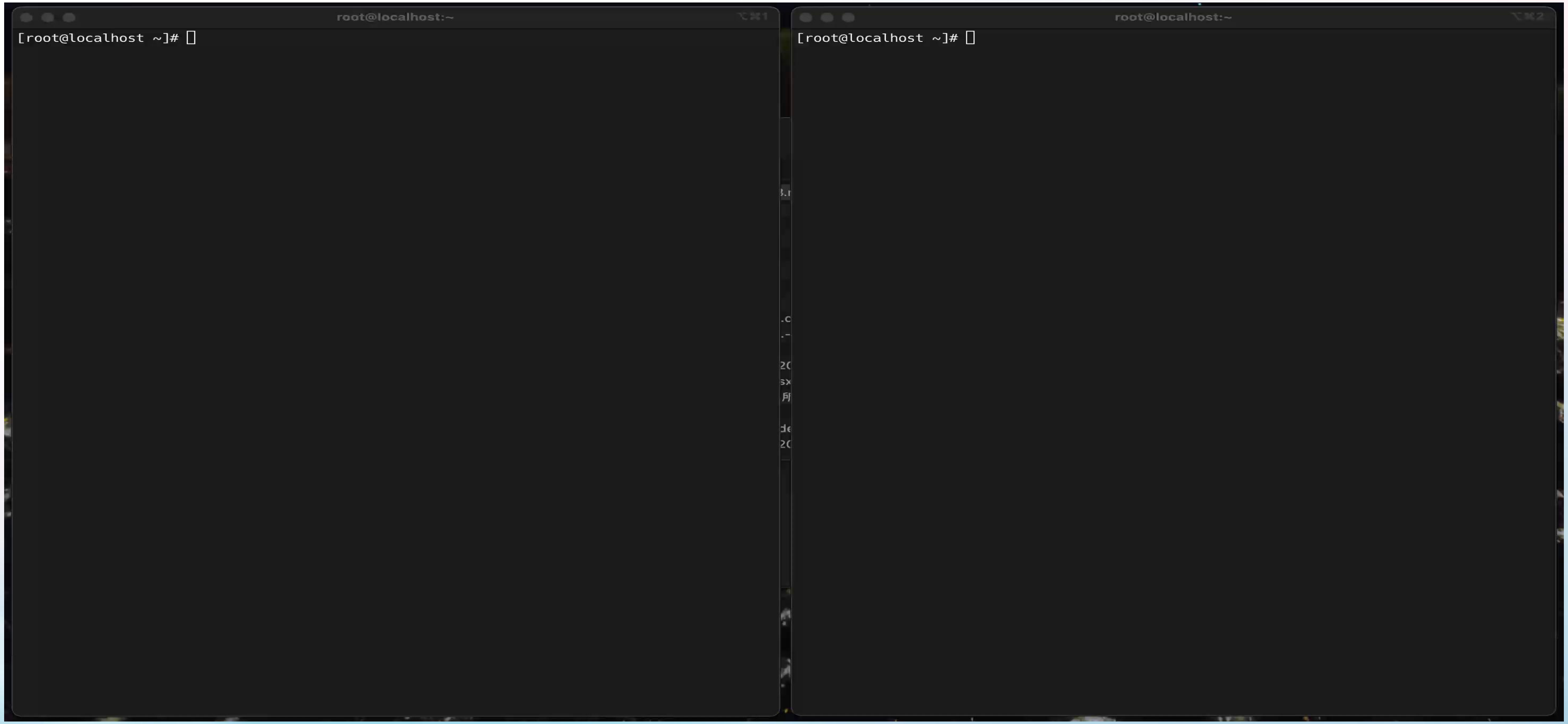
Work Progress — Kata-Containers

We have now successfully launched v3.13.0 Kata-Containers on openEuler RISC-V 24.03 LTS SP1 with its go runtime and QEMU v8.2.0 as its hypervisor.



Demo — Kata-Containers

Invoking Kata with ctr on openEuler 24.03 LTS SP2 to start a ubuntu container image:



Thanks for Listening